

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2683–2713; original printed pp. 89–90; physical source-PDF pp. 79–80; recomposed running pp. 71–72. The compiled reader is the output of the same corrected French edition, not an independent original typescript witness. Line 2714 is blank; the next reduction begins at line 2715 and is excluded.

Assume these results established when X is the spectrum of a regular Noetherian ring A and when $\mathcal{F}$ is the sheaf associated with an A -module of finite projective dimension.

First observe that, if $(X_j)_{j \in J}$ is an open covering of X by open subsets embeddable in a regular scheme, each of the conditions above is equivalent to the conjunction of the analogous conditions obtained by replacing X by X_j , Y by $Y_j = Y \cap X_j$, and $\mathcal{F}$ by $\mathcal{F}|_{X_j}$. Indeed, only the conditions involving $c(x)$ can present a difficulty. Let $x \in U$. If $x \in X_j$, set

$$c_j(x) = \text{codim}(X_j \cap \overline{\{x\}} \cap Y, X_j \cap \overline{\{x\}}).$$

Necessarily $c_j(x) \geq c(x)$. Let $y \in \overline{\{x\}} \cap Y$ be a point that “realizes the codimension,” that is, such that $c(x) = \dim \mathcal{O}_{\overline{\{x\}}, y}$, and let X_j be a member of the covering such that $y \in X_j$. Then $x \in X_j$, and hence $c_j(x) = c(x)$. It follows that condition (a) for the X_j implies condition (a) for X . At this point we have only a partial converse: condition (a) for X implies condition (a) for those X_j such that $c(x) = c_j(x)$, which suffices for our purposes.⁽⁵⁾

⁽⁵⁾*Editor’s note.* In fact, condition (a) for X implies condition (a) for every X_j , as asserted in the original text, but establishing this requires reading the proof below in detail. At this point the implication does not appear formal. For a property (a), (b), or (c), append a subscript J to denote its conjunction over the X_j . The proof *below* establishes $(c_J) \Rightarrow (a_J)$ (this is the chain of implications $(c') \Rightarrow (d) \Rightarrow (a')$). But tautologically $(a) \Rightarrow (c)$, and $(c) \iff (c_J)$; hence $(a) \Rightarrow (a_J)$.